

Reported biomarker–cognition associations do not translate into predictive utility: a pre-registered analysis of NHANES 2011–2014

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Abstract

Background. Multiple studies report associations between blood metals, nutritional biomarkers and cognitive performance among US adults aged 60 and over in the National Health and Nutrition Examination Survey (NHANES) 2011–2014. None isolates the biomarkers’ predictive contribution beyond demographics, with calibration and decision-curve analysis.

Methods. We analysed NHANES 2011–2012 and 2013–2014, restricted to examined participants aged 60 and over. Low cognitive performance was defined as the lowest quartile of a composite z-score from CERAD immediate and delayed recall, animal fluency, and the Digit Symbol Substitution Test. Four nested specifications were pre-specified: demographics (M0); demographics plus five blood metals (M1); demographics plus fifteen nutritional and metabolic biomarkers (M2); and both blocks (M3). A pre-specified secondary baseline (M0+) added eight routinely recorded clinical covariates. Three algorithms (L2-penalized logistic regression, random forest, gradient boosting) were evaluated by 5×10 -fold stratified cross-validation, with preprocessing and tuning fitted inside training folds. The pre-specified primary measure was $\Delta\text{AUC}(\text{M3} - \text{M0})$ with a 2,000-sample paired bootstrap interval; 0.02 was pre-specified as the smallest meaningful improvement. Reporting follows TRIPOD+AI.

Results. Of 3,124 participants with an outcome label, 1,784 had complete predictor data, of whom 431 (24.2%) met the definition of low cognitive performance. Demographics alone reached an AUC of 0.788. $\Delta\text{AUC}(\text{M3} - \text{M0})$ was -0.0019 (95% CI -0.0108 to 0.0071), with the upper bound below the pre-specified 0.02 threshold; neither biomarker block contributed alone. M3 performed worse than the clinical baseline M0+ ($\Delta\text{AUC} -0.0158$, 95% CI -0.0295 to -0.0033) across all three algorithms, and decision-curve analysis showed no net benefit at any threshold. Quantile g-computation on the metal mixture matched demographics alone (AUC 0.7874). The null persisted across all six pre-specified sensitivity analyses, including multiple imputation on the full sample ($n = 3,124$; $\Delta\text{AUC} 0.0015$, 95% CI -0.0030 to 0.0063).

Conclusions. Reported biomarker–cognition associations did not translate into individual-level

predictive utility beyond age, education and income, and a twenty-analyte laboratory panel was 37
outperformed by eight items of routine clinical history. Because the confidence interval excludes 38
the improvement pre-specified as meaningful, this is a precise null rather than an inconclusive one. 39

Keywords: Cognitive performance; biomarkers; blood metals; prediction model; NHANES; older 40
adults; TRIPOD+AI; decision-curve analysis; pre-registration; machine learning 41

Background

Low cognitive performance in older adults is examined routinely against blood metals and nutritional biomarkers in the National Health and Nutrition Examination Survey (NHANES), and panels of these analytes are proposed periodically as adjuncts to demographic risk assessment. Every study making that case for the population and cycles examined here, however, reports only a regression coefficient or an odds ratio. None reports whether the association translates into the ability to identify, at the level of one participant, who has low cognitive performance, and none benchmarks that ability against what is already known from a person's age, sex, race and Hispanic origin, education and family income.

Seven studies span the NHANES 2011–2014 cycles and the population aged 60 years or older analysed here, and none closes that gap. In a preprint, Wang et al. examined blood lead, cadmium, mercury, selenium and manganese in 1,460 participants using linear regression and restricted cubic splines [1]. Lu et al. added serum iron to the same five metals in 2,002 participants using weighted logistic regression [2]. Tang et al. evaluated the same five metals against a dietary inflammatory index in 1,726 participants, using a generalised linear model alongside three mixture-modelling methods: Bayesian kernel machine regression, weighted quantile sum regression and quantile g-computation [3]. Fu et al. examined lead, cadmium, mercury, selenium, copper and zinc jointly in 811 participants using quantile regression and Bayesian kernel machine regression [4]. Song et al. examined the same class of mixture in 1,833 participants stratified by sex [5]. Laouali et al. applied quantile g-computation to lead, cadmium and manganese in 1,777 participants, testing effect modification by two measures of diet quality [6]. Huang and Ren examined an interaction between blood cadmium and dietary omega-6 fatty acid intake in 1,918 participants using logistic regression [7]. Not one of the seven reports discrimination, calibration or cross-validation, and not one compares a biomarker-containing model against a demographics-only baseline.

This is a different question, not a different exposure. Adding one more metal or biomarker to an already saturated association literature is incremental; asking whether any of it predicts, at the

individual level and benchmarked against demographics, is categorical. To our knowledge, that test has not previously been performed on this population, with one recent exception addressed directly here rather than left for the reader to find independently. Ren et al. restricted their sample to NHANES participants aged 60 or older, pooled across the 1999–2000, 2001–2002, 2011–2012 and 2013–2014 cycles, and trained a gradient-boosted model on urinary metals and demographic data to classify what they term “cognitively impaired” status, reporting a cross-validated area under the receiver operating characteristic curve (AUC) of 0.90 [8]. Two things distinguish the present study from theirs. First, NHANES contains no clinical diagnosis of cognitive impairment; the outcome available is a low-performance classification against normed test scores, which we term **low cognitive performance** throughout, and we do not adopt their terminology. Second, and more consequentially for the question this study asks, Ren et al. report the discrimination of one combined model and stop there: their analysis does not decompose that AUC into what demographics alone would achieve and what the metals add beyond it. That decomposition—not combined-model discrimination in isolation—is the quantity a demographics-only baseline is built to measure, and it is what this study reports. A second study, by Nabavi et al., trained a stacking ensemble on heavy-metal biomarkers to predict cognitive impairment across all adults aged 20 and older, reporting an AUC of 0.778, but its population falls outside the age range examined here and it does not report a demographics-only baseline either [9].

The gap between statistical association and predictive utility is a recognised methodological problem beyond this specific exposure-outcome pair, and reporting standards have moved to reflect it: the TRIPOD+AI statement now specifies calibration, decision-curve analysis and internal or external validation as expected components of any prediction-model report, for regression or machine-learning methods alike [10]. None of the association studies above provide any of these. This study does, reported against the TRIPOD+AI checklist in full.

Two limits to this novelty claim are stated here rather than left implicit. The data are not novel: NHANES 2011–2014 is a heavily analysed dataset. The algorithms are not novel: penalised logistic regression, random forests and gradient boosting are standard classifiers. The contribution

is the question, the evaluation framework built to answer it, and the reporting standard applied to it—not a new exposure, outcome, or method.

Objectives. This study estimates how much predictive value blood metals and nutritional biomarkers add, beyond demographics, for identifying low cognitive performance in US adults aged 60 and older, quantified as the change in cross-validated AUC between a demographics-only model and a model additionally containing the biomarker panel, with calibration and decision-curve analysis reported alongside discrimination. This is a model development study with internal validation by repeated cross-validation; no external validation cohort exists for this exposure-outcome pairing, and this is stated as a limitation below. Hypotheses, the primary metric, the threshold for a meaningful difference, and every sensitivity analysis were fixed in writing before any outcome-predictor relationship was examined, and are reported in full in the pre-registration (Additional file).

Methods

Study population and design

We analysed the National Health and Nutrition Examination Survey (NHANES) cycles 2011–2012 and 2013–2014, a complex multistage probability sample of the non-institutionalised civilian US population. Analyses were restricted to participants aged 60 years or older who completed the Mobile Examination Centre visit, the population to whom the cognitive battery was administered. Because two two-year cycles were pooled, the examination weight WTMEC2YR was halved, and the variance strata and primary sampling units were carried through every merge.

The reporting follows TRIPOD+AI. The completed checklist is provided as a supplementary file.

Outcome

Cognitive performance was measured with four scores from three instruments: the Consortium to Establish a Registry for Alzheimer's Disease (CERAD) word-list learning task, scored as the sum of three immediate-recall trials and separately as delayed recall; a one-minute animal fluency task; and the Digit Symbol Substitution Test from the Wechsler Adult Intelligence Scale III.

Each score was standardised within the pooled sample aged 60 and over. A composite was formed as the mean of the available standardised scores, requiring at least three of four to be present, and the lowest quartile of that composite defined **low cognitive performance**. Internal consistency across the four standardised scores was acceptable (Cronbach alpha 0.795); the correlation between CERAD immediate and delayed recall was 0.743, as expected for two scores from one administration, while correlations among the three instruments ranged from 0.40 to 0.51.

The composite is defined against pooled-sample quantiles, so a participant's label depends on the distribution of the whole sample. This is a property of the definition rather than of the prediction procedure: the outcome is explicitly relative, and defining it within folds would make the target itself vary between folds. The choice was pre-specified.

Predictors

Predictors were organised into three blocks fixed in writing before any analysis.

Block 1, demographics. Age, sex, race and Hispanic origin, education, and the family income-to-poverty ratio.

Block 2, blood metals. Lead, cadmium, total mercury, selenium and manganese, measured by inductively coupled plasma mass spectrometry in whole blood, each entered on the natural log scale. Values below the limit of detection are substituted by NHANES with the limit divided by the square root of two and flagged; within the analytic sample this affected 5.2% of cadmium and 5.7% of mercury measurements, and no lead, selenium or manganese measurements.

Block 3, nutritional and metabolic biomarkers. Serum vitamin B12, methylmalonic acid, serum total folate, red blood cell folate, 25-hydroxyvitamin D, glycohaemoglobin (HbA1c), high-density lipoprotein (HDL) cholesterol, total cholesterol, triglycerides, albumin, serum iron, estimated glomerular filtration rate (eGFR; CKD-EPI 2021, without the race coefficient), and three inflammation indices derived from the complete blood count: the neutrophil-to-lymphocyte ratio, the platelet-to-lymphocyte ratio, and the systemic immune-inflammation index. Vitamin B12, methylmalonic acid and the systemic immune-inflammation index were log-transformed.

Covariates. Body mass index, smoking status, average alcoholic drinks per day, the nine-item Patient Health Questionnaire depression score, and self-reported stroke, diabetes, hypertension and high cholesterol. These were not entered in the primary nested comparison, which is stated against demographics, but form a pre-specified secondary baseline described below.

Data preparation

Reserved codes for refusal and non-response were converted to missing on a per-variable basis against each variable's own codebook rather than by a blanket rule, because the applicable codes differ: alcohol quantity uses 777 and 999 over a valid range of 1 to 82, while the cognitive score variables carry no reserved codes at all and a blanket rule would have deleted valid scores of 7 and 9.

Two data properties required specific handling and are noted because both fail silently. First, NHANES SAS transport files encode zero as a denormalised floating-point value rather than as zero, in 330,496 cells across the files used here, with no true zero present anywhere; any equality test against zero therefore matches nothing until this is normalised. Second, alcohol consumption is governed by two nested skip patterns: the screening question refers to any one year of life rather than the past year, and the quantity item is skipped whenever the frequency item is zero. Both indicate zero consumption rather than unknown consumption, and 403 participants were affected.

Derived quantities used in modelling (log transforms, ratios, estimated glomerular filtration rate,

depression score) are row-wise functions with no fitted parameters and were computed before
splitting; all fitted transformations are described below.

Models

Four nested models were specified in advance:

- **M0**, demographics alone, the benchmark against which everything is measured
- **M1**, demographics plus blood metals
- **M2**, demographics plus nutritional and metabolic biomarkers
- **M3**, demographics plus both biomarker blocks

A fifth, pre-specified secondary model **M0+** comprised demographics plus the covariates listed
above, representing information a clinician could record without a laboratory.

The cross-validated performance estimates below come from the nested-CV procedure just de-
scribed, in which each of the 50 folds fits its own pipeline; no single coefficient set exists from
that procedure. For reporting compliance, the identical M3 pipeline was separately refit on the
complete analytic sample, with the regularisation strength selected by the same grid via 10-fold
cross-validation on the full sample (which selected $C = 0.1$, matching the mode across the original
50 folds). Standardised coefficients, odds ratios, and 2,000-resample bootstrap 95% CIs from that
refit are reported in full in a supplementary table.

Three algorithms were fitted to each specification: L2-penalized logistic regression as the pre-
specified primary model class, random forest, and gradient boosting (XGBoost). Multilayer per-
ceptron and k-nearest-neighbour models were excluded in advance, as they were not expected to
be competitive at this sample size.

Analytic sample

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All models were fitted and evaluated on a single locked set of rows, those with every Block 1
to 3 predictor and every covariate observed. Fitting nested models on different samples would
confound model content with sample composition and render the primary contrast uninterpretable.
Nothing was imputed in the primary analysis, so the exposures of interest were never fabricated.

Evaluation

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Models were evaluated by stratified 10-fold cross-validation repeated five times. Every prepro-
cessing step (imputation, standardisation, one-hot encoding) and all hyperparameter selection were
performed inside the training folds using pipeline objects constructed within the cross-validation
loop; no transformation was fitted on data used for evaluation. This was verified by an automated
audit that refits a fold and asserts that the fitted standardisation parameters equal the training-fold
values and not the full-sample values; the audit fails the analysis rather than issuing a warning, and
it passed.

Discrimination was summarised by the area under the receiver operating characteristic curve
(AUC) and the area under the precision-recall curve, overall accuracy being uninformative at
this prevalence. Calibration was summarised by the calibration slope and the calibration-in-the-
large intercept, the latter obtained from the offset model rather than by the common shortcut of
subtracting the mean linear predictor from an intercept-only fit. Clinical utility was assessed by
decision-curve analysis across threshold probabilities from 0.05 to 0.60.

The pre-specified primary outcome measure was ΔAUC **between M3 and M0**, with a 95% con-
fidence interval from 2,000 bootstrap resamples of participants, scoring both models on the same
resampled rows so that the comparison remains paired. A difference of at least 0.02 in AUC was
declared in advance to be the smallest meaningful improvement.

Mixture-modelling benchmark

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To compare the methods used in environmental epidemiology against machine learning on pre-
dictive rather than inferential criteria, quantile g-computation was reimplemented in Python. Ex-
posures are quantised into quartiles using cutpoints learned on the training fold and applied un-
changed to held-out data, matching the behaviour of the R implementation and preserving fold
isolation. The joint mixture effect is the sum of the exposure coefficients, and component weights
are the signed coefficients normalised within direction.

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Bayesian kernel machine regression was excluded in advance: a 2026 simulation study of its pro-
bit implementation reported that only 30 of 431 fits met standard convergence criteria for binary
outcomes, and refitting a Markov chain Monte Carlo model inside 50 folds was not feasible within
the analysis budget. Weighted quantile sum regression was deferred, its package partitioning data
internally in a way that conflicts with an external cross-validation scheme.

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Interpretation

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SHAP values were computed for the gradient-boosted M3 model, with feature rankings recom-
puted across 200 bootstrap resamples to characterise rank stability, since single-run rankings over
correlated predictors are not reliable. Partial dependence was examined for selenium and man-
ganese, the two exposures for which non-monotonic exposure-response was anticipated.

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Sensitivity analyses

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Six analyses were specified in advance: outcome cutoffs at the 10th and 20th percentiles; the con-
tinuous composite as a regression outcome; recoding participants unable to complete the Digit
Symbol Substitution Test as having low performance; restriction to English-language administra-
tion; multiple imputation by chained equations on the full labelled sample with imputation fitted
inside folds; and a leave-one-block-out comparison.

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Software and reproducibility 231

Analyses used Python 3.12.2 with pandas 3.0.3, scikit-learn 1.8.0, XGBoost 3.0.0, SciPy 1.13.0 232
and SHAP 0.52.0. All random seeds were fixed at a single master value and package versions 233
are pinned. The complete pipeline, from file acquisition through figures, is available as seven 234
independently runnable scripts. 235

Results 236

Participants 237

Of 19,931 participants in the two cycles, 3,632 were aged 60 or older and 3,472 had completed the 238
examination visit. Of these, 3,124 had at least three of four cognitive scores and received an out- 239
come label. Requiring complete data on all predictors and covariates yielded the analytic sample 240
of **1,784 participants, of whom 431 (24.2%) met the definition of low cognitive performance.** 241
The weighted population represented was 59.7 million. Participant flow through each stage is 242
shown in Fig. 1. 243

The 2013–2014 cycle contributed proportionally fewer participants (661) than 2011–2012 (1,123), 244
because blood metals in that cycle were measured in a one-half random subsample rather than in 245
all examined participants. 246

Participants excluded for incomplete data were slightly older (mean 70.1 against 69.4 years) and 247
slightly more likely to meet the outcome definition (26.1% against 24.2%), with all differences 248
under three percentage points (Table 2). Participant characteristics for the analytic sample, overall 249
and by outcome status, are shown in Table 1. 250

Discrimination

Demographics alone discriminated moderately well (AUC 0.7884). Adding blood metals (M1, 0.7868), nutritional biomarkers (M2, 0.7895), or both (M3, 0.7865) produced no material change. The clinical baseline M0+ reached 0.8023.

The primary result was $\Delta\text{AUC}(\text{M3} - \text{M0}) = -0.0019$, with a bootstrap 95% confidence interval of $[-0.0108, 0.0071]$ (Fig. 2). The interval includes zero, so the primary hypothesis was not supported. Its upper bound of 0.0071 falls below the 0.02 threshold declared in advance, so the analysis excludes an improvement of the magnitude specified as meaningful rather than merely failing to detect one.

Neither block contributed individually: ΔAUC was -0.0015 $[-0.0052, 0.0023]$ for metals and 0.0011 $[-0.0067, 0.0100]$ for nutritional biomarkers. The question of non-redundancy between blocks does not arise.

The one contrast whose interval excluded zero was negative. **M3 performed worse than the clinical baseline M0+**, by -0.0158 $[-0.0295, -0.0033]$ under logistic regression, -0.0197 $[-0.0338, -0.0055]$ under random forest and -0.0230 $[-0.0385, -0.0063]$ under gradient boosting.

Algorithm comparison

Penalized logistic regression matched or exceeded both flexible learners at every specification. On M3, random forest differed by -0.0059 $[-0.0175, 0.0060]$ and gradient boosting by -0.0086 $[-0.0187, 0.0007]$. The secondary hypothesis that nonlinear learners would outperform penalized regression was not supported.

Logistic regression was also the best calibrated (slope 1.004 on M3, intercept 0.002), whereas random forest was over-dispersed (slope 1.410) and gradient boosting slightly under-dispersed (slope 0.904) (Fig. 3).

Clinical utility

Decision-curve analysis showed no threshold probability at which M3 offered meaningful net benefit over M0 (Fig. 4). M3 exceeded M0 at 21 of 56 thresholds examined, with a maximum gain in net benefit of 0.0059. M0+ exceeded M0 at 52 of 56 thresholds.

Mixture-modelling benchmark

Quantile g-computation applied to the five-metal mixture with demographic adjustment achieved a cross-validated AUC of 0.7874, against 0.7884 for demographics alone and 0.7868 for the same metals entered as individual features. The joint mixture effect was -0.1162 in log-odds per simultaneous one-quartile increase, with cadmium carrying 0.820 of the positive weight and mercury, manganese and selenium loading negatively. A method designed for inferential decomposition of a mixture therefore conferred no predictive advantage over either comparator.

Feature importance

Across 200 bootstrap resamples, education, age and the income-to-poverty ratio occupied the top three ranks with medians of 1, 2 and 3 and appeared in the top five in 100%, 100% and 99% of resamples respectively (Fig. 5). The highest-ranked biomarker was the neutrophil-to-lymphocyte ratio, with a median rank of 5 but a 95% interval spanning ranks 3 to 19 and appearing in the top five in 51% of resamples. Cadmium and selenium had median ranks of 14 and appeared in the top five in 4.5% and 2.5% of resamples. Biomarker rankings were therefore not stable across resamples, while demographic rankings were effectively deterministic.

Partial dependence for selenium and manganese showed no U-shaped exposure-response; predicted probability varied by approximately 0.05 across the observed range of each (Fig. 6).

Sensitivity analyses

The primary finding was unchanged in every pre-specified sensitivity analysis (Fig. 2). $\Delta\text{AUC}(\text{M3} - \text{M0})$ was -0.0019 [$-0.0130, 0.0092$] at the 20th-percentile cutoff, -0.0023 [$-0.0200, 0.0153$] at the 10th, -0.0020 [$-0.0105, 0.0063$] when non-completion of the Digit Symbol Substitution Test was coded as low performance, and -0.0066 [$-0.0159, 0.0028$] when restricted to English-language administration.

Multiple imputation on the full labelled sample of 3,124 gave 0.0015 [$-0.0030, 0.0063$], the narrowest interval obtained in any analysis and still consistent with no difference, indicating that the complete-case restriction did not produce the result.

Leave-one-block-out analysis showed that removing the metals block slightly improved discrimination (-0.0028 [$-0.0051, -0.0004$]), while removing the nutritional block had no effect (-0.0002 [$-0.0081, 0.0078$]).

With the continuous composite as the outcome, cross-validated R-squared rose from 0.3825 for M0 to 0.3908 for M3, a difference of 0.0083. This is the only analysis in which the biomarker blocks contributed in the expected direction, and the magnitude corresponds to under one percent of additional variance explained.

Discussion

Adding a panel of five blood metals and fifteen nutritional and metabolic biomarkers to a demographics-only model did not improve identification of low cognitive performance in US adults aged 60 and older. The primary result, $\Delta\text{AUC}(\text{M3} - \text{M0}) = -0.0019$, 95% confidence interval [$-0.0108, 0.0071$], is a **precise** null rather than an uninformative one: because a difference of 0.02 in AUC was declared in advance as the smallest change worth calling meaningful, the interval's upper bound of 0.0071 rules out an improvement of that magnitude, rather than merely

failing to detect one. That distinction is only available because the threshold was fixed before the
outcome-predictor relationship was examined, and it is the strongest sentence this study can make.

The sharper finding is comparative. The full biomarker panel performed **significantly worse**, not
merely no better, than a baseline of demographics plus eight items of routine clinical history—
body-mass index, smoking status, alcohol use, depressive symptoms, and self-reported stroke, dia-
betes, hypertension and high cholesterol—none of which require a laboratory. That difference held
across all three algorithms, with intervals excluding zero throughout (logistic regression, -0.0158
[-0.0295 , -0.0033]; random forest, -0.0197 [-0.0338 , -0.0055]; gradient boosting, -0.0230
[-0.0385 , -0.0063]). A twenty-analyte panel was outperformed by a clinical questionnaire. This
converts an absence of evidence into a concrete statement about where identifying information
actually resides, and it is the most quotable result in the paper.

The mixture-modelling approach that dominates this literature fares no better when evaluated on
predictive rather than inferential criteria. Quantile g-computation applied to the metals mixture
reached a cross-validated AUC of 0.7874, against 0.7884 for demographics alone: a method built
to decompose the joint effect of correlated exposures conferred no advantage over ignoring the
exposures entirely. Feature-importance rankings compound the caution. Across 200 bootstrap
resamples, the three demographic predictors were essentially deterministic in rank, while every
biomarker's rank moved across a span of fifteen to twenty positions between resamples. A single-
run SHAP ranking over these correlated biomarkers, the form in which such rankings are routinely
reported, is not a reproducible quantity. Taken together with Ren et al.'s report of an AUC of 0.90
for a demographics-plus-metals model on a related population without decomposing that figure
[8], this suggests a broader reporting gap in this literature: strong combined-model performance
and inferentially motivated mixture decompositions are both being reported in place of the specific
quantity that determines whether biomarkers are worth measuring for this purpose, namely their
contribution over demographics alone.

Three misreadings of this result are anticipated and foreclosed here. This study does not show

that any biomarker is unrelated to cognition; the published associations behind the seven studies reviewed in the Background may all be correct as association claims; NHANES is cross-sectional, and nothing here tests, or could test, whether any exposure leads to, causes, or protects against low cognitive performance. Nor does it recommend against measuring these analytes for the clinical purposes for which they are already ordered—vitamin B12 and folate deficiency, glycaemic control, renal function, lipid management—none of which this study evaluated. What it shows is narrower and specific: that translating these published associations into an individual-level prediction of low cognitive performance, in this population, adds nothing measurable once demographics are known.

Several limitations bear on interpretation. The design is cross-sectional, so no temporal or causal claim is possible in either direction. The complete-case analytic sample is mildly non-random—participants excluded for missing data were slightly older and slightly more likely to meet the outcome definition, by under three percentage points on each measure—though the dominant exclusion mechanism, the one-half random subsample used for blood metals in the 2013–2014 cycle, is missing completely at random by design; a sensitivity analysis using multiple imputation on the full labelled sample of 3,124 produced the narrowest interval obtained under any specification (0.0015 [−0.0030, 0.0063]) and remained null, which argues against the sample restriction as an explanation for the result. Age is top-coded at 80 years in the public-use file, compressing the range of the single strongest predictor and likely attenuating, not inflating, any demographic contribution. No external validation cohort exists for this exposure-outcome pairing, so all performance estimates reflect internal, cross-validated discrimination on one population and cycle range; whether the same null holds in a different population is untested. No subgroup fairness audit was pre-specified across race, Hispanic origin or sex, and none is reported. Serum copper and zinc were excluded from the biomarker panel because that assay was run on a one-third subsample in both cycles, and their omission cannot be ruled out as a reason the nutritional block underperforms, though the block’s near-zero contribution held even before that exclusion was applied to any single analyte. Finally, the outcome is defined against pooled-sample quantiles, so an individual

participant's label depends on the distribution of the full sample; this is a declared property of a
relative outcome definition, made explicit in the pre-registration, rather than an artefact discovered
afterward, but it means the label itself is not fixed independent of the sample analysed.

Reported against the TRIPOD+AI checklist, this study is, to our knowledge, the first on this popu-
lation and cycle range to report calibration, decision-curve analysis, and a pre-registered, threshold-
based test of incremental predictive value for these biomarkers. The conclusion is a negative one
on the primary axis, and it is reported as such rather than reframed: seven studies in this literature
report statistically significant associations between these biomarkers and cognition, and this study
finds that translating those associations into identifying which individuals have low cognitive per-
formance adds essentially nothing beyond age, education and income, and measurably less than
eight questions asked without a blood draw.

Conclusions

In this pre-registered, TRIPOD+AI-compliant analysis of NHANES 2011–2014, a twenty-analyte
panel of blood metals and nutritional and metabolic biomarkers added no meaningful predictive
value, beyond age, education and family income, for identifying low cognitive performance in
adults aged 60 and older, and performed significantly worse than a baseline of demographics plus
eight items of routine clinical history. Because the pre-specified 0.02 meaningful-difference thresh-
old was excluded rather than merely not reached, this is a precise null result rather than an incon-
clusive one. These findings describe predictive utility only; they do not bear on, and should not be
read as evidence against, the cross-sectional associations reported elsewhere in this literature.

List of abbreviations

AUC: area under the receiver operating characteristic curve; AUPRC: area under the precision-
recall curve; BMI: body mass index; CERAD: Consortium to Establish a Registry for Alzheimer's

Disease; CI: confidence interval; CKD-EPI: Chronic Kidney Disease Epidemiology Collabora- 394
tion (equation); DPR: Diagnostic and Prognostic Research; eGFR: estimated glomerular filtra- 395
tion rate; HbA1c: glycohaemoglobin; HDL: high-density lipoprotein; IQR: interquartile range; 396
M0/M1/M2/M3/M0+: the four pre-specified nested models and the pre-specified secondary clinical 397
baseline; MICE: multiple imputation by chained equations; NCHS: National Center for Health 398
Statistics; NHANES: National Health and Nutrition Examination Survey; PHQ-9: nine-item Pa- 399
tient Health Questionnaire; SD: standard deviation; SHAP: SHapley Additive exPlanations; TRI- 400
POD+AI: Transparent Reporting of a multivariable prediction model for Individual Prognosis Or 401
Diagnosis, extended for Artificial Intelligence. 402

Participant flow, NHANES 2011-2012 and 2013-2014

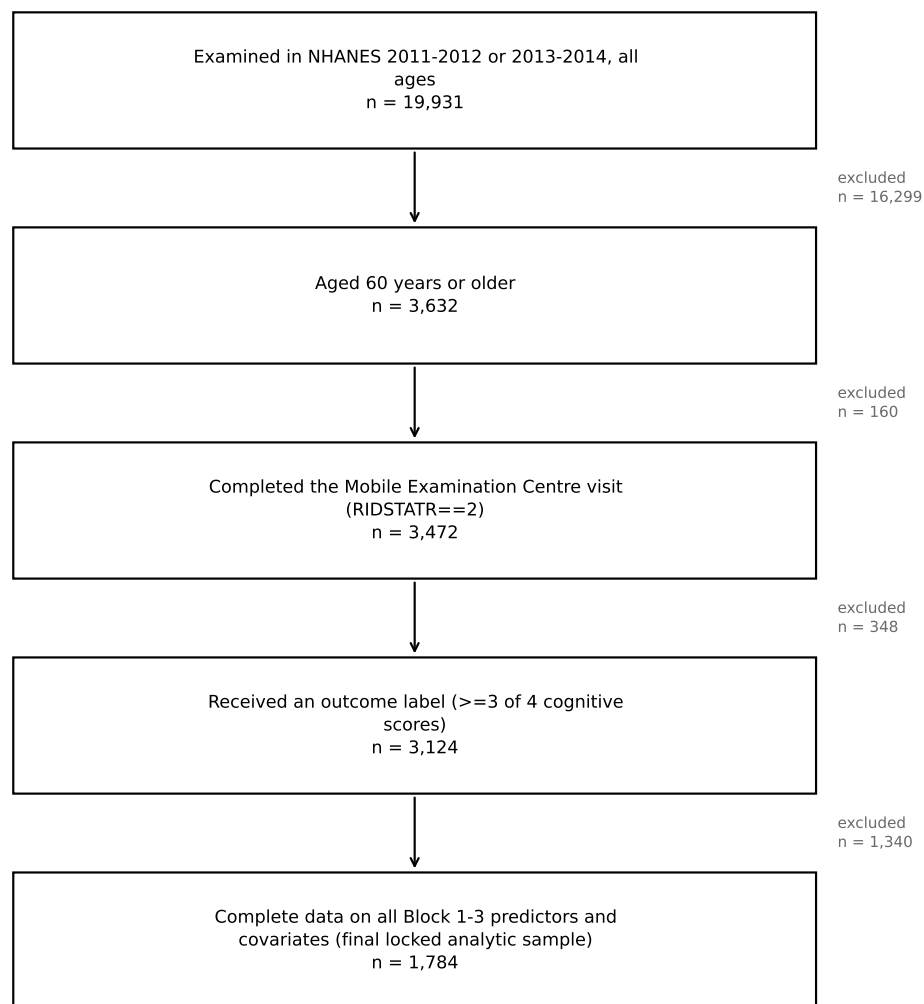


Figure 1: Participant flow from NHANES 2011–2012 and 2013–2014 examined participants to the final locked analytic sample. Reasons for exclusion are given at each stage; the final step’s exclusions (missing predictor data) are not mutually exclusive across blocks.

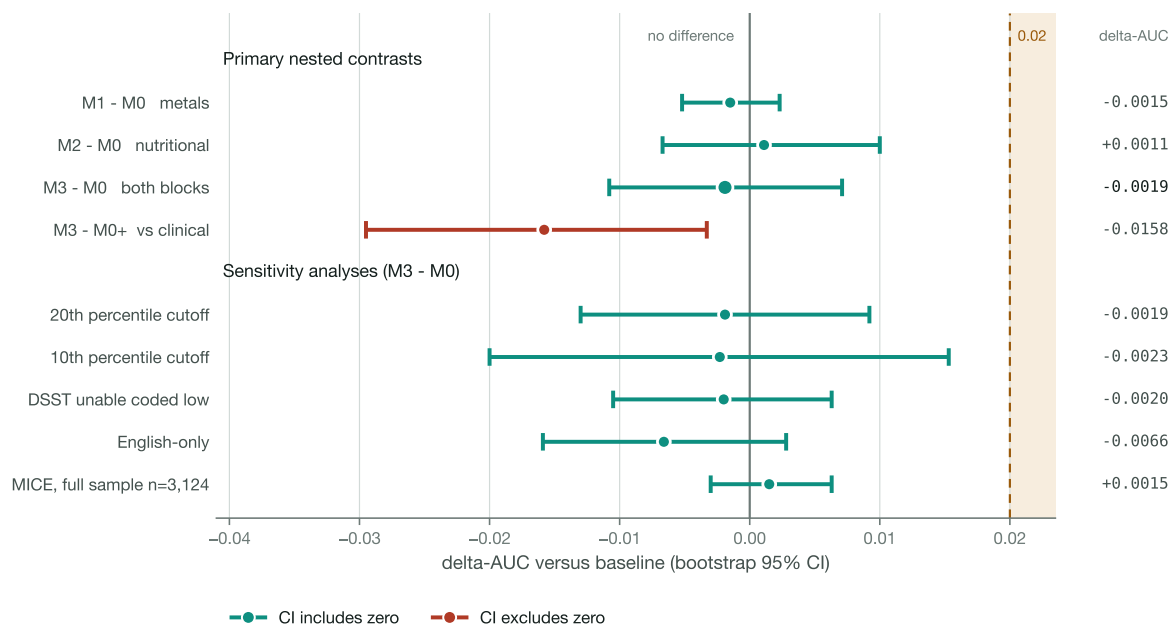


Figure 2: Forest plot of ΔAUC contrasts with bootstrap 95% confidence intervals: the four primary nested contrasts (M1–M0, M2–M0, M3–M0, M3–M0+) and the six pre-specified sensitivity analyses for M3–M0. Red markers indicate intervals excluding zero. The shaded band marks the pre-specified 0.02 meaningful-difference threshold.

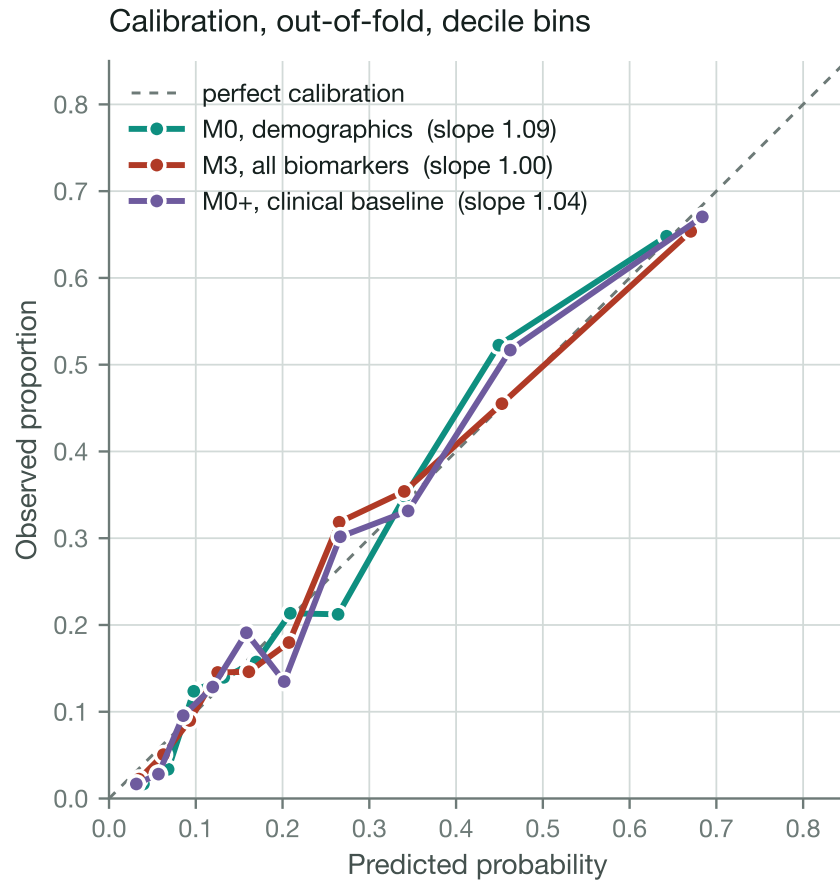


Figure 3: Calibration, out-of-fold, by decile of predicted probability, for M0 (demographics), M3 (all biomarkers) and M0+ (clinical baseline). Calibration slope for each model is given in the legend; the dashed line is perfect calibration.

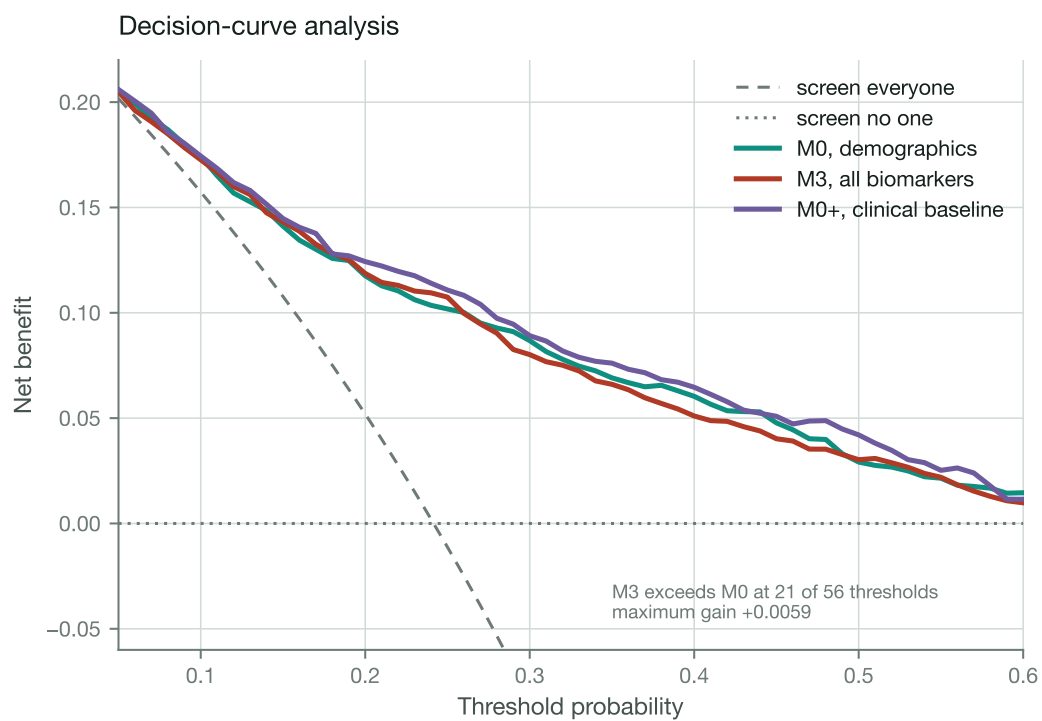


Figure 4: Decision-curve analysis: net benefit against threshold probability for M0, M3 and M0+, against the treat-all and treat-none reference strategies.

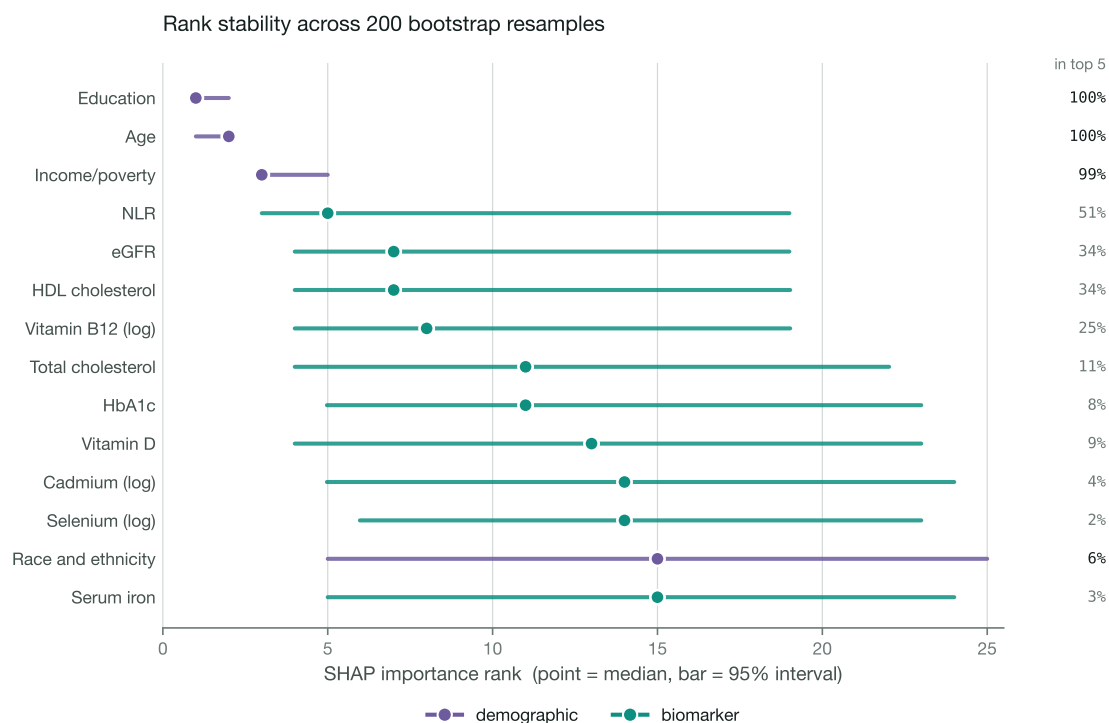


Figure 5: SHAP feature-importance rank stability across 200 bootstrap resamples of the gradient-boosted M3 model. Points are median rank, bars are the 95% interval; the right-hand column gives the percentage of resamples in which each feature appeared in the top 5. Demographic predictors (purple) are essentially deterministic in rank; biomarkers (teal) are not.

No U-shaped exposure-response is present for either essential metal

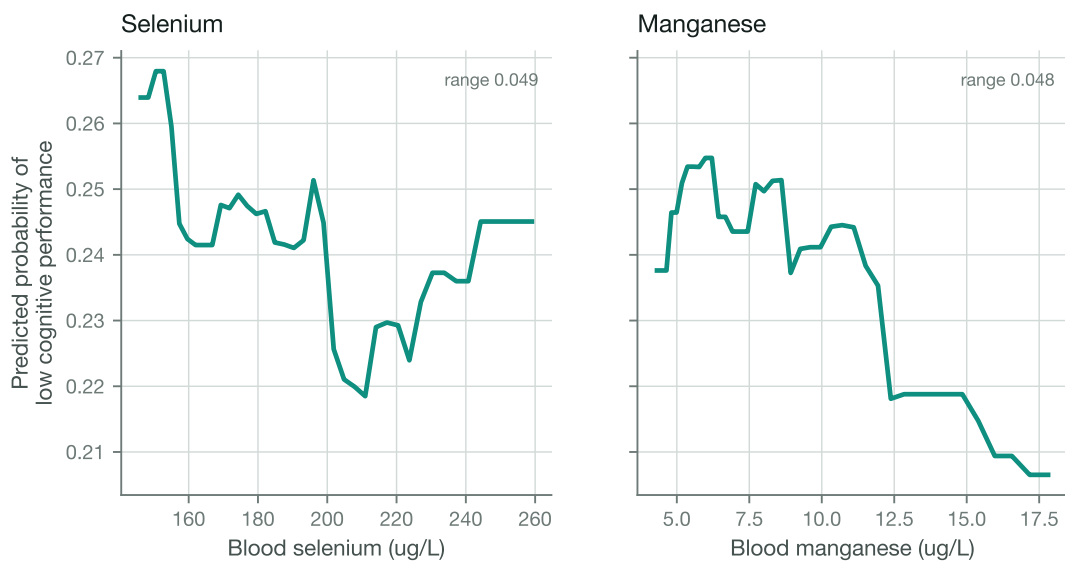


Figure 6: Partial dependence of predicted probability of low cognitive performance on blood selenium and blood manganese concentration (gradient-boosted M3 model), the two exposures for which a non-monotonic (U-shaped) exposure-response was anticipated in advance. Neither shows one; the predicted probability varies by approximately 0.05 across each exposure's observed range.

Table 1: Participant characteristics, analytic sample ($n = 1,784$), overall and by outcome status.

Characteristic	Overall	Low cognitive performance	Not low cognitive performance
<i>n</i>	1784	431	1353
Age, years, mean (SD)	69.37 (6.74)	72.07 (6.94)	68.52 (6.44)
Income-to-poverty ratio, mean (SD)	2.60 (1.60)	1.91 (1.38)	2.82 (1.61)
Blood lead, $\mu\text{g/dL}$, median [IQR]	1.47 [1.01, 2.19]	1.57 [1.03, 2.41]	1.44 [1.01, 2.12]
Blood cadmium, $\mu\text{g/L}$, median [IQR]	0.38 [0.25, 0.61]	0.42 [0.27, 0.69]	0.37 [0.25, 0.58]
Blood total mercury, $\mu\text{g/L}$, median [IQR]	0.93 [0.49, 1.94]	0.75 [0.40, 1.56]	0.99 [0.53, 2.01]
Blood selenium, $\mu\text{g/L}$, median [IQR]	193.09 [177.40, 208.38]	188.35 [173.17, 204.15]	194.23 [179.15, 209.06]
Blood manganese, $\mu\text{g/L}$, median [IQR]	8.71 [6.90, 11.09]	8.35 [6.50, 10.69]	8.80 [7.02, 11.19]
Serum vitamin B12, pmol/L , median [IQR]	410.30 [289.30, 601.45]	421.40 [284.50, 621.80]	408.10 [290.00, 599.30]
Serum methylmalonic acid, $\mu\text{mol/L}$, median [IQR]	163.00 [125.75, 227.00]	181.00 [131.00, 256.00]	160.00 [124.00, 214.00]
Serum total folate, nmol/L , median [IQR]	20.80 [13.60, 31.42]	20.00 [13.05, 32.05]	20.90 [13.70, 31.30]
RBC folate, nmol/L , median [IQR]	543.00 [398.00, 737.00]	530.00 [391.00, 748.50]	547.00 [401.00, 733.00]
25-hydroxyvitamin D, nmol/L , mean (SD)	76.31 (30.44)	72.68 (30.36)	77.46 (30.39)
HbA1c, %, median [IQR]	5.80 [5.50, 6.30]	5.90 [5.50, 6.50]	5.80 [5.50, 6.20]
HDL cholesterol, mg/dL , median [IQR]	52.00 [43.00, 63.00]	49.00 [40.00, 60.00]	52.00 [44.00, 63.00]
Total cholesterol, mg/dL , mean (SD)	192.85 (43.15)	185.58 (41.41)	195.16 (43.45)
Triglycerides, mg/dL , median [IQR]	126.00 [86.00, 187.25]	128.00 [91.00, 182.00]	126.00 [85.00, 189.00]
Serum albumin, g/dL , mean (SD)	4.20 (0.31)	4.15 (0.34)	4.21 (0.29)
Serum iron, $\mu\text{g/dL}$, median [IQR]	80.00 [61.00, 103.00]	76.00 [57.00, 96.00]	82.00 [63.00, 104.00]
eGFR, mL/min/1.73 m^2 , mean (SD)	75.64 (19.85)	69.95 (22.19)	77.45 (18.69)
Neutrophil-to-lymphocyte ratio, median [IQR]	2.11 [1.55, 2.92]	2.35 [1.67, 3.21]	2.05 [1.53, 2.83]
Platelet-to-lymphocyte ratio, median [IQR]	119.80 [94.11, 152.96]	119.05 [89.76, 153.87]	120.00 [95.26, 152.50]
Systemic immune-inflammation index, median [IQR]	458.83 [319.14, 664.17]	480.65 [322.90, 714.20]	452.00 [315.00, 646.00]
Body mass index, kg/m^2 , median [IQR]	28.10 [24.77, 32.40]	27.60 [24.30, 31.90]	28.20 [24.90, 32.50]
Alcoholic drinks per day, median [IQR]	0.00 [0.00, 1.25]	0.00 [0.00, 1.00]	1.00 [0.00, 2.00]
PHQ-9 depression score, median [IQR]	1.00 [0.00, 4.00]	2.00 [0.00, 6.50]	1.00 [0.00, 4.00]
Sex = Male	880 (49.3%)	238 (55.2%)	642 (47.5%)
Sex = Female	904 (50.7%)	193 (44.8%)	711 (52.5%)
Race/ethnicity = Mexican American	140 (7.8%)	46 (10.7%)	94 (6.9%)
Race/ethnicity = Other Hispanic	182 (10.2%)	68 (15.8%)	114 (8.4%)
Race/ethnicity = Non-Hispanic White	863 (48.4%)	165 (38.3%)	698 (51.6%)
Race/ethnicity = Non-Hispanic Black	425 (23.8%)	120 (27.8%)	305 (22.5%)
Race/ethnicity = Non-Hispanic Asian	144 (8.1%)	26 (6.0%)	118 (8.7%)
Race/ethnicity = Other/multiracial	30 (1.7%)	6 (1.4%)	24 (1.8%)
Education = Less than 9th grade	224 (12.6%)	137 (31.8%)	87 (6.4%)
Education = 9–11th grade	219 (12.3%)	79 (18.3%)	140 (10.3%)
Education = High school grad/GED	417 (23.4%)	103 (23.9%)	314 (23.2%)
Education = Some college/AA degree	511 (28.6%)	71 (16.5%)	440 (32.5%)

Table 1 continued

Characteristic	Overall	Low cognitive performance	Not low cognitive performance
Education = College graduate or above	413 (23.2%)	41 (9.5%)	372 (27.5%)
Smoking status = Never	895 (50.2%)	210 (48.7%)	685 (50.6%)
Smoking status = Former	671 (37.6%)	159 (36.9%)	512 (37.8%)
Smoking status = Current	218 (12.2%)	62 (14.4%)	156 (11.5%)
Ever told had a stroke	125 (7.0%)	49 (11.4%)	76 (5.6%)
Ever told had diabetes	490 (27.5%)	154 (35.7%)	336 (24.8%)
Ever told had high blood pressure	1111 (62.3%)	302 (70.1%)	809 (59.8%)
Ever told had high cholesterol	997 (55.9%)	234 (54.3%)	763 (56.4%)

Table 2: Included versus excluded participants, among the 3,124 outcome-eligible participants.

Characteristic	Included (<i>n</i> = 1,784)	Excluded (<i>n</i> = 1,340)
Age, years, mean (SD)	69.37 (6.74)	70.10 (6.94)
Income-to-poverty ratio, mean (SD)	2.60 (1.60)	2.50 (1.59)
PHQ-9 depression score, median [IQR]	1.00 [0.00, 4.00]	2.00 [0.00, 5.00]
Female, %	50.7%	52.6%
Low cognitive performance, %	24.2%	26.1%
Diabetes, %	27.5%	30.0%

Declarations

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Ethics approval and consent to participate

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NHANES protocol and procedures were approved by the National Center for Health Statistics 405
(NCHS) Ethics Review Board (formerly the NCHS Research Ethics Review Board), and written 406
informed consent was obtained from all participants by NCHS prior to data collection. This study 407
is a secondary analysis of de-identified, publicly available data and involved no additional contact 408
with participants. Both authors confirmed that their institutions (Davidson College and the Uni- 409
versity of North Carolina at Chapel Hill) impose no additional IRB requirement for this secondary 410
analysis. 411

Consent for publication

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Not applicable. No individual person's data, images, or identifiable information are presented; all 413
results are reported at the level of aggregate statistics on a de-identified public dataset. 414

Availability of data and materials

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NHANES data are publicly available from the CDC at www.cdc.gov and are not redistributed 416
here; the download script rebuilds the raw dataset from the original CDC URLs. The analysis 417
code is available at <https://github.com/NeilP211/nhanes-ieee>, under version control, with 418
every random seed fixed and every package version pinned. 419

Competing interests

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The authors declare that they have no competing interests. 421

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Authors' contributions

425

S.K. and N.P. conceived the study and jointly decided the analytic sample definition, the nested
model specification, and the pre-registration. S.K. acquired the raw data. N.P. curated the data
and developed and ran the primary modelling pipeline: cohort construction, outcome construction,
the nested M0–M3/M0+ models across three algorithms, cross-validated evaluation, interpretation
(SHAP, partial dependence), and the six sensitivity analyses. S.K. conducted the literature review
and citation verification, and conducted the participant flow-diagram, Table 1, and full model co-
efficient analyses. N.P. wrote the original draft of the Methods and Results; S.K. wrote the original
draft of the Abstract, Introduction, and Discussion. Both authors reviewed and edited the complete
manuscript and approved the final version. S.K. and N.P. contributed equally to this work and share
first authorship.

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